

Multimodal Fake News Detection using CLIP and Retrieval-Augmented LLMs

Sneha Soni, Shruti Yadav, Abhilipsha Behera, Shochisha, Vanshika Verma, and
Rishabh Kaushal

Department of Information Technology
Indira Gandhi Delhi Technical University for Women, Delhi, India
{sneha063btaiml23, shruti061btaiml23, abhilipsha003btaiml23,
shochisha058btaiml23, vanshika075btaiml23, rishabhkaushal}@igdtuw.ac.in

Abstract. Social media has made it really easy for fake news to spread especially when it comes to posts that have both text and images. A lot of these posts use pictures but with fake or misleading captions, which makes it hard to figure out what is real and what is not. To catch news we need systems that can look at how the text and images work together and understand what is going on in the real world. The problem with AI systems is that they only look at the text and pictures they are given and they do not check with the real world. This means we cannot always trust what they say.

In this project we are making a system to detect news that uses a few different tools: BERT, Vision Transformer and CLIP. We also have a part that writes captions for images and a special pipeline that helps us find evidence from news articles and Wikipedia. We use a language model to analyze all the evidence we find. Then we put all the information together to make a decision about whether something is fake news or not.

Our system is designed to be accurate and to give us results that we can trust. Overall this system is useful to find whether the news is real or fake.

1 Author Contributions

Divide the tasks as discussed in guidelines among the team members. Put in the table below who is doing which task.

Table 1: Task Distribution Among Team Members

Student Name	Enrollment Number	Task Assigned
Shochisha	05801192023	Data Pre-processing, BERT, ViT
Sneha Soni	06301192023	Related Work, Methodology, CLIP, Pipeline, Interpretation, Discussion
Shruti	06101192023	Introduction, Captioning, API, RAG-based LLM, Pipeline, Methodology, Limitations
Abhilipsha	00301192023	Fusion layer, Classification, Methodology, Results
Vanshika Verma	07501192023	Abstract, Conclusion, Future Work, References

2 Introduction

In recent years, the widespread circulation of fake news has increased rapidly because of multimedia platforms like TikTok and YouTube. This made it more difficult to differentiate which information is real and which one is fake [9].

Earlier, the detection of fake news is based on unimodal approaches like text-based or image-based classification [10]. This makes traditional methods insufficient since it only relies on textual and visual information [5]. But recently research is shifted towards multimodal techniques which successfully combines both textual and visual features together as it will lead to the accurate detection [10].

Additionally, Large Language Models (LLM) has also drew attention in detecting fake news because of their strong reasoning abilities. But to effectively use them for fake news detection is still challenging [10].

Therefore, due to the large amount of shared information, it makes it difficult to accurately verify the information or the news. The verification of information is crucial, since spread of fake news can influence the opinions of people, create panic, lead to misinformation to critical areas such as health, social issues and politics etc. Further false information can be used to manipulate users, disrupt systems etc.

To address the problem, we designed a multimodal fake news detection using CLIP and Retrieved-Augmented LLMs. This not only combines Textual and visual information but also add retrieved evidence used by Language model to predict real/fake/uncertain, give justification and confidence score. All these things are further combined using fusion layer to generate 3-way and 2-way classification.

3 Related Work (M2)

Fake news detection is been studied lot over time. Previously, the focus was on to find that the text whether its real or fake but with time, researchers notice that images and text are sometimes does not aligned with each other. Strengths and weaknesses are mentioned in Table 2.

Table 2: Strengths and Weaknesses of Related Works.

Authors	Strengths	Weakness
Nakamura et al. [1]	Large model, fine-grained labeling in three different ways	labeling by distant supervision gives noisy labels
Zhao et. al. [2]	Cross-modal graph that captures relation between image and text	Fixed threshold
Tomar et. al. [3]	simple, fast and not too long dataset used	fails when fake news looks real
Hu et. al. [4]	LLMs are used for explanations	LLM alone is not accurate and enough
Wang et. al. [5]	multiple models are combined for better accuracy	very costly
Wu et. al. [6]	focus on actual meaning rather than writing	training is complex and expensive
Qi et. al. [7]	strong dataset	No clear reasoning
Hu et. al. [8]	multiple models are combined and works well for shorter video	weakness
Yan et. al. [9]	generate explanations even without ground truth	complex and slow
Hu et. al. [10]	combine graph and LLM using mask	ignores low confidence data

Nakamura et al. [1] introduced Fakeddit, a large Reddit dataset of over one lakh samples with different labeling like 2-way, 3-way and 6-way combining text and image data. This labeling was done through distant supervision method which is automatic labelling of data without any human intervention and it is found out after human annotation that those labels were not so accurate. Zhao et. al. [2] proposed a new solution, FCN-LP, where tweets are connected to each other in a graph using similarities from CLIP features. Labels are then marked between shared posts. This approach uses a fixed threshold and basic propagation, which makes it prone to error when post which are not similar is connected. Tomar et. al. [3] uses a simpler approach where using CLIP, he checks the image and text alignment. If the image is fake and serves the meaning similar to text then CLIP score would be high and creates a problem when fake content will be marked as real. As we move forward LLM came into the picture, Hu et al. [4] uses LLMs like GPT-3.5 for fake news detection, and used them as to generate an explanation for their detection but it does not give an accurate answer. Later on, Wang et al. [5] introduces FND-LLM, which take multiple models like BERT, ViT, CLIP, and GPT-4 and combine them. It is then given to co-attention transformer and MMoE network which helps in achieving improvements over Weibo, GossipCop, and PolitiFact dataset. But frequent use of GPT-4 API calls makes it costly. Wu et al. [6] proposed SheepDog, which trains both on normal and sensational news. It is used to focus mostly on content rather than writing. Qi et al. [7] proposed FakeSV and SV-FEND, which uses information like what is

posted and what are the comments i.e. it studies the people participation. Later, Hu et al. [8] proposed MAGE-fend model specially for short videos, which takes text, images, audio, and LLM-reasoning, and it performs better than previous models proposed. Recently, Yan et al. [9] proposed DIFND, which uses diffusion model to generate explanations even when no ground truth is available. It uses LLM in order to give better reasoning. Hu et al. [10] proposed GLPN-LLM, which uses LLM pseudo labels into a cross-modal graph and use masking mechanism in order to prevent label leakage, and is used on Twitter, PHEME, and Weibo dataset.

Multimodal fake news detection has improved a lot but some problems still exist such as many systems that use LLMs, like FND-LLM and MAGE-fend, depends only on the knowledge they already know or that is feeded into their system, they do not check real, up-to-date information and can sometimes give incorrect explanations. Also, some methods like GLPN-LLM ignore low-confidence completely. Another issue is that most of the systems only give a final result i.e. whether the information is real or fake, without explaining how much confident they are in their predictions.

To solve all these problems, we proposed a simple and efficient multimodal system. We use CLIP to check whether the image and text match or mismatch by giving a similarity score, BERT is used to understand the text meaning, and ViT is used to extract image features. In addition, we used a RAG-based LLM that retrieves real-time information from sources like Wikipedia and news APIs, so that model can verify facts instead of relying only on its own internal knowledge. All these features are then combined and passed through a neural network to make the final prediction which we called main confidence. We have also calculated an extra confidence score using CLIP similarity, LLM confidence, and retrieved information. This makes our system more transparent and reliable.

4 Data Details

The dataset used in this project is the FakeReddit dataset which contains Reddit post and annotations for misinformation classification. Original dataset link: <https://github.com/cvdfoundation/google-landmark?tab=readme-ov-file>

4.1 Dataset Description (M1)

The FakeReddit Dataset is used to detect fake news and misinformation on Reddit. It includes both text-based and metadata features from Reddit post. Data collection Process:- It was collected using:

- Dataset was collected using the Pushshift.io API.
- Dataset has content from 19 march 2008 to 24 october 2019.
- The subreddit used is PhotoshopBattles.

Dataset has title, images, user comments and meta data like score, author username upvote and downvote. Dataset Features

- Number of rows:- 26,679
- Data type: Mixed type(textual,numeric,boolean)
- Target Feature: 2 way and 3 way label

Table 3: Details of Data Attributes.

Data Attributes	Data Type	Brief Explanation
image_url	text	URL of the images
title	text	Original post title
2_way_label	numeric	Binary classification label
3_way_label	numeric	Fake,real and other label

4.2 Data Pre-processing (M1-M2)

Data need to be pre-processed for this project work. The Preprocessing Code is <https://colab.research.google.com/drive/1gfVwqA2ecvQ5eo0-AziJ7XQnAt-7yL0y#scrollTo=ZYXbbJIF97yU>(**google colab link**) that is performed on the original dataset.

- Data Quality Issues: There were various data quality problems in the dataset, such as missing values, empty image links, irrelevant observations like hasImage=False and duplicates. The issues were resolved by: Deleting rows with missing image_link or title keeping only relevant images like hasImage=True, Removing duplicates based on image_link and columns. By cleaning the data the resulting size of the dataset became 26679 rows.
- Data Fusion: The dataset is multimodal i.e. text and images. Text is extracted from "title" column and given to BERT. Images are transformed using ViT. Both text and images are processed in CLIP. Data fusion will happen by combining text and image information to perform classification.
- Data Sampling: No sampling techniques were used as the size of the dataset was small. Data was filtered only after cleaning and kept natural label distribution.
- Synthetic data: No synthetic data generation methods were used. Project requires only real data. small original data.
- Fairness: Dataset is not biased towards any specific classes, such as real, fake, satire, etc.
- Robustness: Since distant supervision, so the labels can be noisy or not accurate.

5 Pipeline

The pipeline of this proposed architecture is shown in Figure 1. The input consists of a news image and its claim. Text feature is extracted using BERT, image

features using ViT, CLIP is used for text-image alignment and LLM score is extracted by semantic understanding. All extracted features are then concatenated and is converted into single feature vector, which is passed through a neural network and main confidence is calculated with final classification.

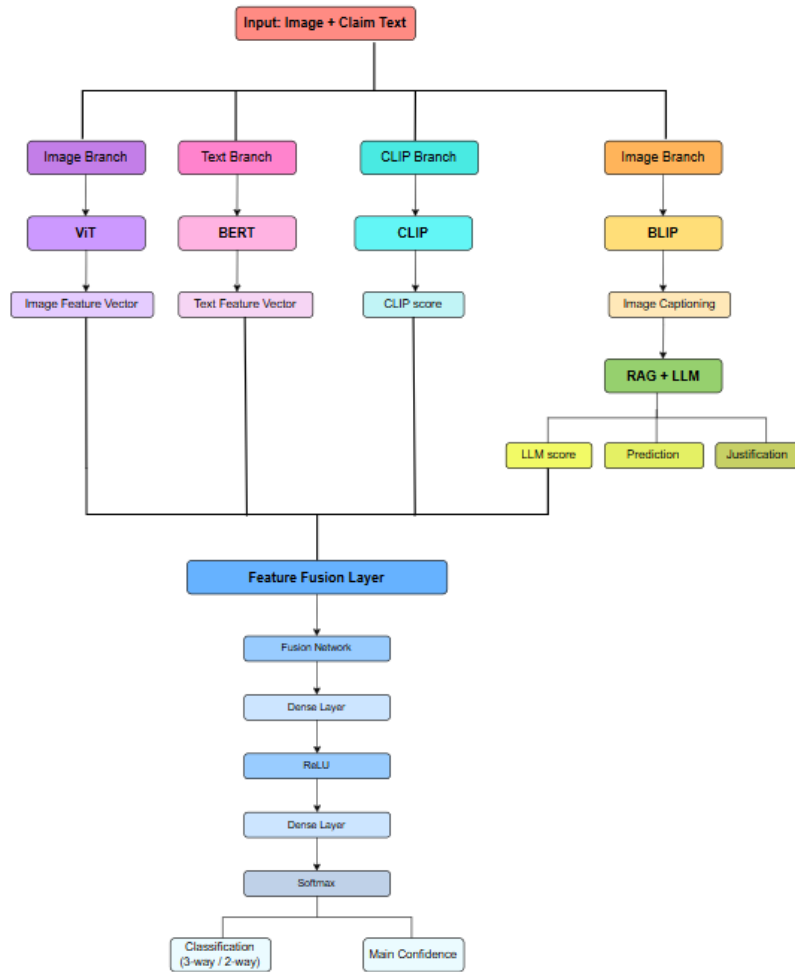


Fig. 1: Proposed Multimodal Fake News Detection Pipeline

6 Methodology (M3)

BERT(Bidirectional encoder representations from) is the tool used for feature extraction for text embedding. This language model uses transformer technology to understand the context of the word in the sentence, analyzing the text by looking left and right(bidirectionally) simultaneously. Here, the BERT algorithm is used in "title" attribute of the dataset. Firstly, the text is tokenized through a pretrained BERT tokenizer, after which it will be passed through BERT to generate text embedding. The resulting vector will be of 768 dimension.

For visual feature extraction, the Vision Transformer(ViT) model is used. This differs from conventional CNN-based models since images are divided into patches then patches are converted into vectors or embeddings and then embeddings are transferred to transformer and then transformer learns the global dependencies of the image. Image feature vector is of size 768-dimension.

CLIP (Contrastive Language-Image Pretraining) developed by OpenAI that tells how similar an image and text are with each other. It calculates the semantic relationship between image and text. CLIP is used here to analyze the relation between news image and caption to tell whether it's real or misleading. CLIP uses image encoder which converts an image into feature vector and text encoder converts the text into another feature vector. The similarity score is calculated using cosine similarity, which is used as a measure of alignment. Both text and image encoder obtain a 512-dimensional feature vector. We have calculated CLIP similarity score using features. The formula for Cosine similarity is given below:

$$\text{cosine similarity} = \frac{\mathbf{f}_{\text{image}} \cdot \mathbf{f}_{\text{text}}}{\|\mathbf{f}_{\text{image}}\| \|\mathbf{f}_{\text{text}}\|}$$

The Retrieval-Augmented Generation(RAG) is an AI framework that is designed to improve Large-Language model (LLM) accuracy by retrieving data from external, trusted knowledge bases before generating a response. It helps in reducing hallucinations, helps in verifying claims by providing evidences. Therefore, the LLM generates an answer based on both its trained knowledge and the retrieved data.

For our system, the RAG based LLM module is designed to verify the news claims using both image and textual information. It combines claim, image (which is converted into textual description), evidence retrieval and LLM to generate prediction along with justification and confidence score.

Now, let's go through the complete pipeline. The system takes two inputs from the dataset: the textual claim and the associated image url.

The dataset is loaded from a CSV file and basic preprocessing is done to ensure consistency. The relevant fields, including the claim text and image url are then extracted for the use in the pipeline.

Using HTTP request, the image is fetched from the url. The retrieved image is in binary format, which is then converted into file-like object. The Python Imaging Library (PIL) is used to open and process this data as an image. After this the image is further converted into RGB format. Now this processed image is then used for generating captions using BLIP model.

The BLIP model, which is a pretrained image captioning model is used to generate textual information from the input image.

To support the verification process, external evidences are retrieved from NewsAPI providing real- time data and WikipediaAPI providing background knowledge. Now based on the claim, it fetches recent news article titles and descriptions along with short wikipedia summary. It helps in providing factual context for the claim.

Since not all retrieved information are relevant, a Sentence Transformer model is used to filter the evidences. Using Sentence Transformer, the claim and evidence are converted into embeddings and then cosine similarity is computed i.e., computing semantic similarity between the claim and evidence chunks. Then the most relevant evidences are selected based on the similarity score for the final reasoning. It helps in reducing noise and improving the quality of input provided to the LLM.

Now, the claim, image caption and selected evidences are passed to the language models input. Here, Gemini 2.5 Flash Lite model is used as a language model. Based on its trained knowledge and the selected evidences, the model analyzes whether the claim is supported, contradicted, or unclear then it generates a structured response.

Here, a structured prompt is used, which provide instructions and clearly defines output format. It allows model to follow structure, which ensures consistency in the responses and makes it easier to extract results.

The final output includes prediction i.e., whether the claim is real/fake/uncertain, along with justification which is explaining the reasoning, and a confidence score representing the certainty of the decision.

Further to ensure the robustness of the system, error handling mechanisms are implemented throughout the system. In cases where the image url is invalid, empty or cannot be retrieved, a default response is used. API errors are managed using exception handling. Retry logic is applied to handle temporary failures and to implement attempt request multiple times. To avoid exceeding API rate limits, a delay is introduced between consecutive requests. This helps in preventing request failures as API has limit and too many request can cause error.

After extracting all the required features from different models, these image features, text features, CLIP similarity score, and LLM-based features were combined together into a single input vector and that finally is passed to fusion model. Each features have their own meaning such as image features represent visual information, text features tells semantic meaning, CLIP score provides image-text alignment, and LLM score tells the confidence of its prediction. By combining them, the model is able to learn relationships of multiple models used. The final concatenated vector serves as the input to the fusion model.

$$F_{\text{final}} = \text{Concat}(F_{\text{text}}, F_{\text{image}}, F_{\text{CLIP}}, F_{\text{LLM}})$$

To effectively combining these different features and learn complex interactions between cross modal, we built a Gated Multimodal Fusion Network (GatedFusionNet). This modal introduces an flexible gating mechanism that

measures the contribution and its importance of each feature. The gate function act as a soft attention mechanism that enables model to minimise the influence of irrelevant information and highlights important elements. This is essential for combining features which is coming from distinct distribution.

In GatedFusionNet network consists of 3 main modules

1. Projection layer – That converts the combined vectors into a high dimensional space
2. Gating module – It computes the per dimensional specific weight vector using a sigmoid activation function
3. Classifier – It generates final class probabilities.

The gating operation involves calculating represented features by a gate vector, followed by two-layer classifier with batch normalization and dropout to prevent overfitting.

To train the proposed GatedFusionNet, the dataset was split using stratified shuffling (80% training, 20% testing). To address this class imbalance in the 3-way task (6 classes initially which is merged into 3 classes), the training dataset was balanced using random oversampling. The features were standardized with effective scaling method. The training process uses 150 epochs a batch size of 16 and cross entropy loss with smoothed labels which is applied at a rate of 0.1. we use AdamW optimizer set up with learning rate of 3×10^{-4} and weight decay of 0.01, and a one cycle learning rate organiser with a maximum learning rate of 1×10^{-3} . Following training, softmax was used to generate test predictions.

Initially the previously extracted feature matrix and its associated labels were loaded from numpy files. The file `X_features.npy` stored the combined multimodal vectors for every sample, and `y_2.npy` provides binary labels which differentiate between “Fake” and “Real”. And `y_6.npy` contains original six class labels. These arrays provide the data needed for the following preprocessing, training, and evaluation steps.

Continuing from the loaded data, the original six class distribution was analyzed, which exposed a major class imbalance (counts: 62, 9, 22, 3, 2, 2). To support more stable training, rare classes (1, 3, 4, 5) were merged into single class labelled as class “other”. Following this, a mapping dictionary $\{0:0, 2:1, 6:2\}$ was used to convert the combined labels into a continuous three-class system (0, 1, 2). Later, the `y6` array was applied in the three-way classification task, and its even distribution was confirmed through an additional output. Combining these steps lessened the number of classes and reduced the effect of categories with sparse data.

Following the confirmed training and evaluation protocols the binary classification task was performed by the `train_model` function, using the two-class labels `y2` and without oversampling. The model displayed 80.00% accuracy on the test set and an average prediction confidence of 0.7590.

For the three-class classification task (which was created by merging the original six classes into three), the `train_model` function was called with `num_classes` set to 3 and `use_oversampling` set to `True`. Oversampling was required because the class distribution continued to be imbalanced (counts: 62, 22, 16) even after

merging. If it is not balanced, the model would show biasness towards the majority class (Class0). Randomly oversampling the training set balanced the class frequencies to 49 each, allowing the model to learn additional different features among all categories. The model reached an accuracy of 75.00% and had a mean confidence of 0.7700.

Now we proceeded to train the we proceeded to train the GatedFusionNet on the complete dataset (all 100 samples) in order to obtain predicted labels and confidence scores for every instance. After this step it is essential when the aim is to enhance the original CSV with model-generated annotations for further analysis or semi-supervised learning. For the three-class task, the same oversampling method (RandomOverSampler) was use on the entire training dataset to save class balance. The model undergoes training for 150 epochs with hyperparameters: RobustScaler for feature normalization, a batch size of 16, AdamW optimizer, OneCycleLR scheduler, and label-smoothing for cross-entropy loss. Following training, the model made predictions on all 100 samples, generating predicted class labels (0=Class0, 1=Class2, 2=Other) along with their confidence scores (like confidence_score_2way & confidence_score_3way. These results were subsequently incorporated as new columns into the original CSV, leading to an enhanced dataset (final_fusion_dataset_with_predictions.csv).

7 Result

7.1 Visualization of Model Performance

Training Accuracy Curves (2-way vs 3-way)

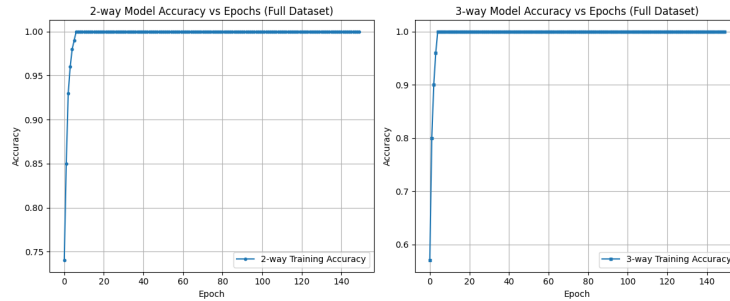


Fig. 2: Training accuracy of 2-way and 3-way models on the full dataset.

The training accuracy curves for both the 2-way and 3-way models achieved 100% within the initial 20 to 30 epochs and maintained the perfect accuracy for the next 150 epochs. This shows that GatedFusionNet has enough capacity to keep the complete training set of 100 samples. The rapid joining can be assigned to the AdamW optimizer and the OneCycleLR scheduler. The unusual results on

the training set contrast with accuracy levels on the test set (80% for 2-way and 75% for 3-way), which points to overfitting; the model has picked up on patterns specific to the dataset rather than more in-general relevant characteristics. The imbalance indicates that additional data contains more improved regularization rate could improve generalization.

Binary (2-Way) Classification

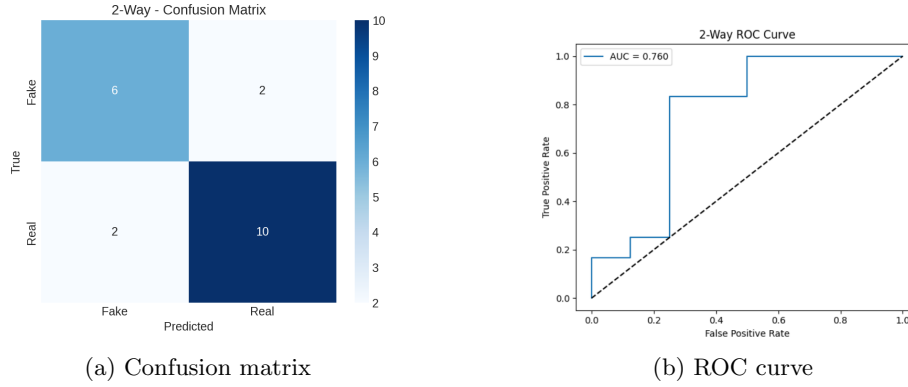


Fig. 3: Performance visualizations for the binary classification task.

For 2-way task: The confusion matrix for the binary classification task it indicates that from 8 true “fake” samples, 6 were accurately detected while 2 were inaccurately classified as “Real”. Out of the 12 true “Real” samples, 10 were accurately identified and 2 were incorrectly labelled as “fake”. It indicates that model performs slightly better on the “Real” class matching with the precision and recall values mentioned previously. The ROC curves shows the balance between true positive rate and the false positive rate at various classification thresholds. The area under the curve (AUC) is 0.760, shows moderate level of classification ability. The curve increases upward toward the top-left corner not extremely steeply, indicating that the model is able of attaining a moderate true positive rate while keeping false positive rate low. Overall model shows good performance but not optimal discriminative performance.

Three-Way Classification

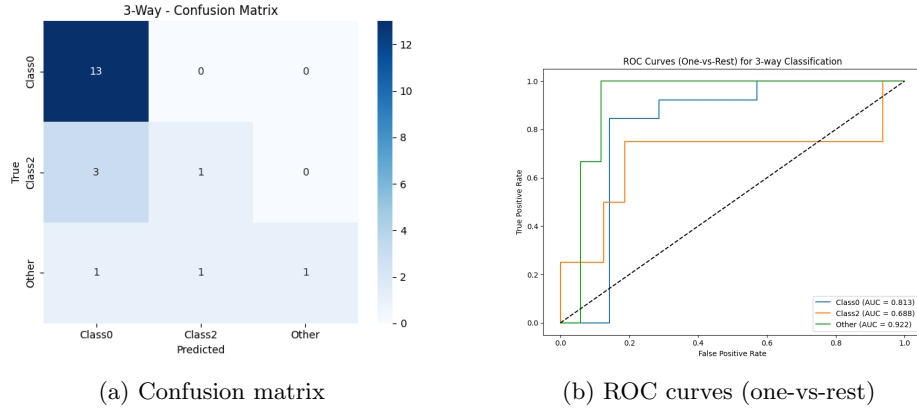


Fig. 4: Performance visualizations for the three-way classification task.

3-way classification task: The confusion matrix for the 3-way task shows that all the 13 true class 0 samples are correctly classified. In class 2 just 1 of the 4 samples are correctly classified where as other three were incorrectly classified as class 0. In the “other” category which was created by merging classes 1,3,4 and 5, the model exhibit minimal effectiveness, attaining just 1 correct prediction from 3 samples, with the remaining samples are misclassified as class 0 and class 2. This shows that model continues to have difficulty in differentiating withing the “other” category and class 2 likely because of class imbalance and the mixed nature of the combined class, even though the performance for Class 0 has improved.

The ROC curves shows the balance between the true positive rate and the false positive rate in a one-versus-rest scenario. The class 0 shows the AUC of 0.813 indicating the effectiveness of classification capabilities. Class 2 shows moderate AUC of 0.688 showing less separability. In comparison “other” category shows the AUC of 0.922 which indicates excellent ability to differentiate between the class through ranking. The results shows that models shows best performance for class 0 and has good ranking ability for “other” category, but somewhere it struggles with minority classes, especially on class 2.

7.2 Classification report 2-way and 3-way

Table 4: Binary classification report (Fake vs. Real).

Class	Precision	Recall	F1-Score	Support
Fake	0.75	0.75	0.75	8
Real	0.83	0.83	0.83	12
Accuracy		0.80		20
Macro Avg	0.79	0.79	0.79	20
Weighted Avg	0.80	0.80	0.80	20

Mean Confidence (2-way): 0.7590

Table 5: Three-way classification report (Class0, Class2, Other).

Class	Precision	Recall	F1-Score	Support
Class0	0.76	1.00	0.87	13
Class2	0.50	0.25	0.33	4
Other	1.00	0.33	0.50	3
Accuracy		0.75		20
Macro Avg	0.75	0.53	0.57	20
Weighted Avg	0.75	0.75	0.71	20

Mean Confidence (3-way): 0.7700

7.3 Examples

This section presents examples to show the effectiveness of our model.



Fig. 5: Input Image

Claim: ig nobel win for alpine goat man

CLIP Score: 0.32

LLM Confidence: 0.9

Final Model Confidence: 0.96

Prediction (Final Output): Real

Evidence: A parody of the Nobel Prizes, the Ig Nobel Prizes are awarded each year in mid-September, around the time the recipients of the genuine Nobel Prizes are announced, for ten achievements that "first make people laugh, and then make them think". Commenting on the 2006 awards, Marc Abrahams, editor of *Annals of Improbable Research* and co-sponsor of the awards, said that "[t]he prizes are intended to celebrate the unusual, honor the imaginative, and spur people's interest in science, medicine, and technology".

Justification: The evidence describes the Ig Nobel Prizes as awards for achievements that "first make people laugh, and then make them think," and that they celebrate the unusual. The claim of an "ig nobel win for alpine goat man" aligns with the nature of these parody awards, suggesting it is a plausible, albeit humorous, event. The image of a man petting a goat further supports the "goat man" aspect of the claim, making the entire scenario consistent with the Ig Nobel Prize criteria.



Fig. 6: Input Image

Claim: the sun hitting my orange soda just right

CLIP Score: 0.28

LLM Confidence: 0.6

LLM Prediction: Uncertain

Final Model Confidence: 0.96

Prediction (Final Output): Fake

Evidence: No external evidence found. Analyze based on claim: the sun hitting my orange soda just right

Justification: The claim is a subjective observation about how light is interacting with a drink. The image shows a person holding a glass of liquid, which is consistent with the claim. However, there is no objective factual evidence to verify or disprove the specific aesthetic quality described. It's a personal experience, not a statement of fact.

8 Interpretation

To understand what are the models used in the proposed fusion model have contributed, an ablation study was performed. In this study, the model was trained multiple times by removing specific features and observing the change in accuracy.

- **All Features Included (Image + Text + CLIP + LLM):** The model was trained using all the features and this achieved accuracy of **80%**, which serves as the baseline performance of the proposed fusion model.
- **Without using CLIP features:** Next, we removed CLIP-based similarity scores while keeping all other features intact. The model achieved a higher accuracy of **85%**. This indicates that CLIP features were not contributing positively and may have introduced noise or redundancy. It can also be interpreted that the information provided by CLIP may be already captured by other features such as image and text embeddings.
- **Without LLM Features:** Next, we removed LLM scores while keeping all other features intact. The model accuracy dropped to **75%**, which is lower than the baseline. This tell us the LLM played a very important role in improving model accuracy.

After ablation study, we get to know that The LLM features are highly influential, as when we removed them it leads to a decrease in accuracy from 80% to 75%. The CLIP features are less useful as when we removed them the accuracy increases from 80% to 85%. This may be due to feature redundancy where other features such as image and text embeddings may already captured the information.

9 Discussion (M3)

- The results shows that the proposed multimodal fusion approach has achieved strong performance with an accuracy of 80% in the 2-way classification and 75% in the 6-way classification. The ablation study shows that not all features contribute equally to the model. In particular, LLM-based features improved the performance.
- As we observed from the results, that without CLIP score accuracy has been increased, this tells us that maybe due to redundancy CLIP score is not giving that contribution, and the small dataset size may have limited the model’s ability to effectively learn from CLIP features.
- On the other hand, removing LLM scores results in decrease of accuracy, LLM has captured the semantic information from image caption and given claim which enhances the model’s ability to distinguish between classes.
- As we had gone through previous works, it has been observed that some work heavily relied on CLIP. Our work shows that CLIP may not always improve performance in all scenarios. Comparative study has been shown in Table 6.
- While some research work report higher accuracy around 86%, they have computed on larger datasets or with different experiment environment. Whereas, our model has achieved 80% of accuracy on a small dataset, which shows the robustness of our model under constrained conditions.

Table 6: Ablation Study Results

Configuration	Accuracy (%)
Full Model (All Features)	80
w/o CLIP Features	85
w/o LLM Features	75

10 Limitations

- Limited Dataset Size: Because of the external constraint, the system is evaluated for 100 datapoints. However, this sample size is enough to verify that the pipeline is functioning correctly. Hence, we have designed a proof of concept system. For more reliable generalization, a larger dataset would be required.
- API Rate Limit: It is one the major limitation of the RAG pipeline. We have used Gemini API for a language model but it don’t allow unlimited usage, only certain number of request per minute/hour/day. Due to this within the project timeline, we are only able to process 100 rows, which affected the scalability. In future it can be extended with higher API quotas.

- Image Captioning Accuracy (BLIP): Another limitation is, the system uses pre-trained image captioning model which converts image into textual descriptions. Now for complex or low quality image (i.e., complex scenes, blurry image, too many objects, misleading visuals, etc) the generated caption can be inaccurate or vague. In result, it can affect the LLM Reasoning or prediction.
- CLIP features require sufficient data to show meaningful impact. With a small dataset, the model is unable to effectively learn patterns from CLIP scores.
- It was found out that the image-text alignment scores produced by CLIP were restricted to a narrow range (approximately 0 to 0.5) and did not exceed this range.
- Image and text embeddings already captured semantic information and as a result, CLIP similarity may become redundant and does not provide any additional useful information.
- Due to extremely small dataset i.e., 100 datapoints, it may results in high overfitting risk.
- To address the class imbalance, random oversampling is used so that minority classes samples can be increased. This may lead to overfitting as existing data is being duplicated rather than introducing new information.
- Loss of rare class information: The less frequent class is merged into single 'other' category. While this helps in simplifying classification tasks but results in loss of meaningful distinctions between rare classes.
- Another Limitation is lack of external validation. The system has been evaluated only on the available dataset because of the external constraint (i.e., API Rate Limit.)

11 Conclusion (M3)

This paper presents a multimodal fake news detection framework that combines text, images, and external knowledge to make predictions more reliable. It uses BERT to understand text, Vision Transformer (ViT) to extract visual features, and CLIP to connect image and text information. Along with this, a RAG-based large language model helps bring in real-time knowledge, so the system isn't just relying on what it learned earlier but can actually check information in context. Another important part of the model is the gated multimodal fusion mechanism, which helps it decide how much importance to give to each type of input instead of treating everything equally.

The results show that the model performs well on both binary and multi-class classification tasks, even when the data is imbalanced. Methods like stratified splitting, oversampling, and feature standardization help make the training process more stable and improve how well the model works on new data. On top of that, using external evidence through APIs and semantic similarity filtering helps the model support its predictions with relevant information and confidence scores, making the outputs more trustworthy and easier to understand.

The ablation study makes it clear that not every part of the model contributes equally. Features coming from the large language model have a strong impact on improving accuracy, while CLIP-based features contribute less and may need further improvement. Overall, this study reinforces the value of evaluating individual feature contributions within multimodal architectures and offers meaningful guidance for refining model design and improving performance in future research.

12 Future Work (M3)

The data set definitely needs to be larger and more diverse. Now, it is very limited, which means that the model might just be memorizing patterns instead of understanding them. If we train it on more type of data it will be much better to handle real world situations.

Another issue is data imbalance, some of the classes don't have many examples so the model doesn't really learn them, this can make biased predictions towards the more frequent classes. Using smart approaches like improved sampling techniques or loss function can help fix and make the model treat all the classes same.

The way features are combined can be more improved. At the moment it is quite basic and it might be limiting the models performance. We could use more methods which are advanced that can focus on the most important features.

Fine tuning models like CLIP and LLMs specifically for this task makes a big difference, right now they are being used as they are without adapting the problem, with proper fine tuning it can capture more relevant patterns which can improve the accuracy.

It is very important to test the model on real-world data. The real world data is very messy and unpredictable, evaluating the model in those conditions will give a much clearer idea of how reliable it actually is.

Resources

The dataset, extracted features, and supporting files used in this work are available at:

<https://drive.google.com/drive/folders/1rfaEVGvN7Ki0asoR51thyID6TreUEopf?usp=sharing>

References

1. K. Nakamura, S. Levy, and W. Y. Wang, “Fakeddit: A new multimodal benchmark dataset for fine-grained fake news detection,” in *Proceedings of the Twelfth Language Resources and Evaluation Conference*, N. Calzolari, F. Béchet, P. Blache, K. Choukri, C. Cieri, T. Declerck, S. Goggi, H. Isahara, B. Maegaard, J. Mariani, H. Mazo, A. Moreno, J. Odijk, and S. Piperidis, Eds. Marseille, France: European Language Resources Association, May 2020, pp. 6149–6157. [Online]. Available: <https://aclanthology.org/2020.lrec-1.755/>
2. W. Zhao, Y. Nakashima, H. Chen, and N. Babaguchi, “Enhancing fake news detection in social media via label propagation on cross-modal tweet graph,” in *Proceedings of the 31st ACM International Conference on Multimedia*, ser. MM ’23. New York, NY, USA: Association for Computing Machinery, 2023, p. 2400–2408. [Online]. Available: <https://doi.org/10.1145/3581783.3612086>
3. S. Tomar, P. Sharma, D. Saini, and R. Kaushal, “Fake news detection by contrastive language-image pretraining using thresholding,” in *2023 2nd International Conference on Vision Towards Emerging Trends in Communication and Networking Technologies (ViTECoN)*, 2023, pp. 1–5.
4. B. Hu, Q. Sheng, J. Cao, Y. Shi, Y. Li, D. Wang, and P. Qi, “Bad actor, good advisor: Exploring the role of large language models in fake news detection,” *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 20, pp. 22 105–22 113, 2024. [Online]. Available: <https://ojs.aaai.org/index.php/AAAI/article/view/30214>
5. J. Wang, Z. Zhu, C. Liu, R. Li, and X. Wu, “Llm-enhanced multimodal detection of fake news,” *PLOS ONE*, vol. 19, no. 10, p. e0312240, 2024. [Online]. Available: <https://doi.org/10.1371/journal.pone.0312240>
6. J. Wu, J. Guo, and B. Hooi, “Fake news in sheep’s clothing: Robust fake news detection against llm-empowered style attacks,” in *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, ser. KDD ’24. New York, NY, USA: Association for Computing Machinery, 2024, p. 3367–3378. [Online]. Available: <https://doi.org/10.1145/3637528.3671977>
7. P. Qi, Y. Bu, J. Cao, W. Ji, R. Shui, J. Xiao, D. Wang, and T.-S. Chua, “Fakesv: A multimodal benchmark with rich social context for fake news detection,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2023, pp. 14 444–14 452. [Online]. Available: <https://ojs.aaai.org/index.php/AAAI/article/view/26689>
8. L. Hu, Z. Wang, J. Zhu, Y. Hu, and X. Wang, “Mage-fend: Multimodal adaptive fusion with guidance from llm expertise for fake news detection on short video platforms,” *Knowledge-Based Systems*, vol. 329, p. 114298, 2025. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0950705125013395>
9. K. Yan, M. Liu, Y. Liu, R. Fu, Z. Wen, J. Tao, and X. Liu, “Debunk and infer: Multimodal fake news detection via diffusion-generated evidence and llm reasoning,” 2025.
10. S. Hu, J. Hu, and H. Zhang, “Synergizing llms with global label propagation for multimodal fake news detection,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, 2025, pp. 1426–1440. [Online]. Available: <https://aclanthology.org/2025.acl-long.72/>
11. J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, “Bert: Pre-training of deep bidirectional transformers for language understanding,” in *Proceedings of the 2019 Conference of the North American Chapter of the Association*

- for *Computational Linguistics*, 2019, pp. 4171–4186. [Online]. Available: <https://aclanthology.org/N19-1423/>
12. A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly *et al.*, “An image is worth 16x16 words: Transformers for image recognition at scale,” *arXiv preprint arXiv:2010.11929*, 2020. [Online]. Available: <https://arxiv.org/abs/2010.11929>
 13. J. Li, D. Li, C. Xiong, and S. Hoi, “Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation,” in *Proceedings of the International Conference on Machine Learning (ICML)*, 2022, pp. 12 888–12 900. [Online]. Available: <https://arxiv.org/abs/2201.12086>
 14. N. Reimers and I. Gurevych, “Sentence-bert: Sentence embeddings using siamese bert-networks,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP)*, 2019, pp. 3982–3992. [Online]. Available: <https://aclanthology.org/D19-1410/>
 15. G. Team, R. Anil, S. Borgeaud, J.-B. Alayrac, J. Yu, R. Soricut, J. Schalkwyk, A. M. Dai, A. Hauth, K. Millican *et al.*, “Gemini: A family of highly capable multimodal models,” *arXiv preprint arXiv:2312.11805*, 2023. [Online]. Available: <https://arxiv.org/abs/2312.11805>
 16. NewsAPI.ai, “Newsapi.ai – news search and analysis platform,” 2024. [Online]. Available: <https://newsapi.ai/>
 17. Wikipedia Contributors, “Wikipedia, the free encyclopedia,” 2024. [Online]. Available: <https://www.wikipedia.org>
 18. Hugging Face, “Transformers library documentation,” 2024. [Online]. Available: <https://huggingface.co/docs/transformers>
 19. Sentence Transformers, “Sentence-transformers documentation,” 2024. [Online]. Available: <https://www.sbert.net>
 20. Python Software Foundation, “Python language reference,” 2024. [Online]. Available: <https://www.python.org>
 21. Estarek, “Model fusion,” 2024. [Online]. Available: <https://medium.com/@estarek007/model-fusion-336d5828e729>
 22. GeeksforGeeks, “Fusion learning - the one shot federated learning,” 2025. [Online]. Available: <https://www.geeksforgeeks.org/machine-learning/fusion-learning-the-one-shot-federated-learning/>
 23. T. Meng, X. Jing, Z. Yan, and W. Pedrycz, “A survey on machine learning for data fusion,” *Information Fusion*, vol. 57, pp. 115–129, 2020. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S1566253519303902>
 24. Sapien, “Mastering multimodal data fusion: How to combine text, images, audio more,” 2025. [Online]. Available: <https://www.sapien.io/blog/mastering-multimodal-data-fusion>
 25. M. Al-alshaqi, D. B. Rawat, and C. Liu, “A bert-based multimodal framework for enhanced fake news detection using text and image data fusion,” *Computers*, vol. 14, no. 6, p. 237, 2025. [Online]. Available: <https://www.mdpi.com/2073-431X/14/6/237>
 26. S. Alonso-Bartolome and I. Segura-Bedmar, “Multimodal fake news detection,” *arXiv preprint arXiv:2112.04831*, 2021.